

MAJOR PROJECT REPORT

REGRESSION AND CLASSIFICATION BASED MACHINE LEARNING ANALYSIS

August 2026 Batch

Prepared By

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Programme: Data Science & Business Analytics (DSBA)

Project Title:

House Price Prediction | Loan Approval Prediction | Admission Accuracy Improvement

Submission Date: 30th September 2026

Acknowledgement

I express my sincere gratitude to the faculty and mentors associated with the Data Science & Business Analytics programme for providing the guidance, learning resources and project framework required to complete this Major Project. I am also thankful for the practical exposure to regression, classification, feature engineering, model evaluation and accuracy-improvement techniques.

The project provided an opportunity to apply classroom concepts to structured datasets and to build reproducible machine-learning workflows in Python. The work was completed with an emphasis on data preprocessing, fair model comparison, measurable evaluation and clear interpretation of results.

Abstract

This Major Project implements three machine-learning studies covering regression and classification. Project 1 develops a house-price prediction system using Bangalore housing data and compares Linear Regression, Decision Tree and Random Forest models using RMSE, MAE and R^2 . Project 2 develops a loan-approval classification system using applicant demographic, financial, credit and property-related attributes and compares Logistic Regression, Decision Tree, Random Forest and Support Vector Machine models using Accuracy, Precision, Recall, F1 Score and ROC-AUC. Project 3 revisits an Admission Problem and establishes a Linear Regression baseline before applying feature engineering, ensemble methods, polynomial expansion, Ridge regularization and hyperparameter tuning.

The experiments produced measurable results. In Project 1, Random Forest achieved an RMSE of 59.498, MAE of 27.197 and R^2 of 0.7155. In Project 2, Logistic Regression achieved 86.18% accuracy, 90.81% F1 Score and 0.8486 ROC-AUC, while Random Forest recorded the highest precision of 85.39%. In Project 3, Tuned Polynomial Ridge reduced RMSE from 0.0679 to 0.0677 and increased R^2 from 0.8212 to 0.8224, corresponding to a 0.34% RMSE reduction.

Keywords: Machine Learning, Regression, Classification, Feature Engineering, Random Forest, Logistic Regression, Polynomial Ridge, Model Evaluation

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Chapter 1 – Introduction

Machine learning provides a practical approach for extracting patterns from structured data and using those patterns to support prediction and analysis. Regression models estimate continuous outcomes, while classification models predict categorical outcomes. In both cases, the quality of the final system depends not only on the algorithm but also on data cleaning, feature representation, evaluation methodology and interpretation.

This Major Project applies these principles through three related studies. The first study addresses property-price estimation, where the target is a continuous house price. The second study addresses loan approval, where the target is a binary approval status. The third study focuses on improving an existing admission-probability regression model through feature engineering and advanced model selection.

The project follows an end-to-end workflow: dataset acquisition, exploratory analysis, preprocessing, feature engineering, train-test splitting, model training, evaluation, comparison and interpretation. A consistent emphasis is placed on using held-out test data and reporting actual observed metrics rather than assuming that a particular algorithm will perform best.

Project Objectives

- Build reproducible regression and classification pipelines using Python and scikit-learn.
- Explore and preprocess real-world structured datasets.
- Compare multiple machine-learning algorithms using appropriate evaluation metrics.
- Identify influential features and translate model results into practical insights.
- Improve the Admission Problem baseline using polynomial features, Ridge regularization and hyperparameter tuning.
- Document limitations and avoid target leakage or artificial temporal features.

Chapter 2 – Basic Concepts / Literature Review

2.1 Regression

Regression predicts a continuous numerical target. Linear Regression models the relationship between predictors and the target using a linear combination of coefficients. Tree-based regressors such as Decision Trees and Random Forests can represent nonlinear relationships and interactions.

The main regression metrics used in this project are RMSE, MAE, MAPE and R^2 . RMSE penalizes larger errors more strongly, MAE represents average absolute error, MAPE expresses error as a percentage when appropriate, and R^2 represents the proportion of target variance explained by the model.

2.2 Classification

Classification predicts discrete classes. Logistic Regression estimates class probabilities, Decision Trees learn rule-based partitions, Random Forest combines multiple decision trees, and Support Vector Machines identify a separating boundary using a maximum-margin principle.

Accuracy measures overall correct predictions. Precision measures the proportion of predicted positive cases that are actually positive. Recall measures the proportion of actual positive cases correctly identified. F1 Score is the harmonic mean of precision and recall. ROC-AUC summarizes ranking performance across classification thresholds.

2.3 Feature Engineering and Model Improvement

Feature engineering transforms raw variables into representations that may expose useful relationships to a model. In Project 3, polynomial feature expansion was combined with Ridge regularization. Ridge adds an L2 penalty to reduce excessive coefficient magnitude and helps control overfitting when polynomial features increase the number of predictors.

2.4 Time Series and LSTM

Time-series analysis relies on ordered observations, where lag variables and rolling statistics can capture temporal dependence. LSTM is a recurrent neural-network architecture designed to preserve useful information over longer sequences through forget, input and output gates. The Admission dataset used in Project 3 is cross-sectional rather than genuinely time-indexed; therefore, arbitrary row-order lag features were not introduced because they would create artificial temporal relationships.

Chapter 3 – Problem Statement / Requirement Specifications

The Major Project guideline requires three studies spanning regression and classification. Each study must include data exploration, preprocessing, model development and evaluation. The final work must provide an updated notebook and a comparative analysis of model performance.

3.1 Project Planning

- Acquire or load the specified public dataset.
- Inspect data types, missing values, duplicates, distributions and target variables.
- Clean and transform the data while preserving the meaning of variables.
- Split data into training and test sets and keep evaluation data unseen during model fitting.
- Train the required baseline and comparison models.
- Evaluate models using the metrics specified for each project.
- Perform feature importance and practical interpretation.
- Document results, limitations and conclusions.

3.2 Requirements

Project	Target	Required Models	Primary Metrics
Project 1	House Price	Linear Regression, Decision Tree, Random Forest	RMSE, MAE, R^2
Project 2	Loan Status	Logistic Regression, Decision Tree, Random Forest, SVM	Accuracy, Precision, Recall, F1, ROC-AUC
Project 3	Chance of Admit	Baseline + advanced/tuned regression	RMSE, MAE, MAPE, R^2

3.3 Design Constraints

The project is implemented in Python using pandas, NumPy, Matplotlib, Seaborn and scikit-learn. The notebooks are designed for execution in Google Colab or a compatible Jupyter environment. Model comparisons use the same held-out test data within each project. The Project 3 Admission dataset used for the executed experiment is the standard Admission_Predict.csv-style Graduate Admissions dataset; the project guideline itself specifies that the class dataset should be reused, so the exact class file should be substituted if it differs.

Chapter 4 – Implementation

4.1 Project 1 – Predicting House Prices in India

The first project uses a Bangalore/Bengaluru housing dataset containing property characteristics such as area type, location, size, total square footage, bathrooms and price. The workflow included missing-value handling, parsing of total square footage ranges, categorical encoding, outlier treatment and feature engineering.

A price-per-square-foot variable was used for exploratory analysis and outlier investigation but was deliberately excluded from predictive features because it is directly derived from the target price and would create target leakage. Additional engineered variables included square footage per BHK and bathrooms per BHK.

The cleaned dataset produced 11,825 usable observations for the train-test workflow, with 9,460 training observations and 2,365 testing observations.

4.1.1 Model Results

Model	RMSE	MAE	R ²
Linear Regression	66.377	31.472	0.6459
Decision Tree	67.973	30.047	0.6286
Random Forest	59.498	27.197	0.7155

The Random Forest Regressor produced the lowest RMSE among the three evaluated models, with RMSE 59.498, MAE 27.197 and R² 0.7155. Linear Regression achieved R² 0.6459, while the Decision Tree achieved R² 0.6286.

4.1.2 Feature Importance and Business Insight

Rank	Feature	Aggregated Importance
1	total_sqft	0.722
2	Location	0.091
3	sqft_per_bhk	0.087
4	area_type	0.031
5	bathrooms	0.022

The feature-importance analysis indicates that total property area was the dominant feature in the Random Forest model, followed by location and square footage per BHK. These findings suggest that property size and location-related information play an important role in price estimation within this dataset.

4.2 Project 2 – Predicting Loan Approval

The second project uses the standard Loan Prediction / Loan Eligibility dataset with applicant demographic, financial, credit and property-related variables. The target variable Loan_Status is mapped from Y/N to approved/not approved. Missing values are imputed, categorical variables are one-hot encoded, numerical features are scaled where appropriate, and class imbalance is considered.

A derived TotalIncome feature was created by combining applicant and coapplicant income. Exploratory analysis included approval rates by demographic and financial categories, distributions, correlation analysis and credit-history relationships.

4.2.1 Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.8618	0.8400	0.9882	0.9081	0.8486
Decision Tree	0.8537	0.8526	0.9529	0.9000	0.8296
Random Forest	0.8211	0.8539	0.8941	0.8736	0.8260
Support Vector Machine	0.8455	0.8438	0.9529	0.8950	0.8316

Logistic Regression achieved the highest accuracy at 0.8618, the highest recall at 0.9882, the highest F1 Score at 0.9081 and the highest ROC-AUC at 0.8486 among the evaluated models. Random Forest achieved the highest precision at 0.8539. These results illustrate that model performance depends on the evaluation metric and the operational objective.

4.2.2 Feature Importance and Banking Interpretation

Rank	Feature	Approx. Aggregated RF Importance
1	Credit_History	0.316
2	ApplicantIncome	0.132

3	TotalIncome	0.128
4	LoanAmount	0.125
5	CoapplicantIncome	0.079

Credit history was the most influential feature in the Random Forest analysis, followed by applicant income, total income, loan amount and coapplicant income. Feature importance indicates how the trained model used variables for prediction; it should not be interpreted as proof of a causal relationship.

4.3 Project 3 – Accuracy Improvement Challenge

The third project establishes a transparent Linear Regression baseline for the Admission Problem and then evaluates feature engineering and advanced regression methods. The executed experiment uses the standard Graduate Admissions structure with GRE Score, TOEFL Score, University Rating, SOP, LOR, CGPA, Research and Chance of Admit.

Meaningful engineered features included an academic test index, profile strength, academic profile, research-university interaction and CGPA-GRE interaction. Because the dataset is cross-sectional rather than time-indexed, artificial lag and rolling features were not created from row order.

4.3.1 Baseline and Improved Results

Model	RMSE	MAE	MAPE (%)	R ²
Tuned Polynomial Ridge	0.0677	0.0476	8.44	0.8224
Polynomial Ridge	0.0679	0.0481	8.54	0.8213
Baseline Linear Regression	0.0679	0.0480	8.51	0.8212
Extra Trees + Features	0.0705	0.0492	8.72	0.8077
Tuned Random Forest + Features	0.0706	0.0497	8.77	0.8071
Tuned Extra Trees + Features	0.0710	0.0498	8.85	0.8048
Random Forest + Features	0.0716	0.0501	8.80	0.8015

Expanded Tuned Gradient Boosting + Features	0.0726	0.0511	9.07	0.7958
Gradient Boosting + Features	0.0727	0.0498	8.80	0.7951
Tuned Gradient Boosting + Features	0.0733	0.0507	8.99	0.7919

The final expanded comparison identified Tuned Polynomial Ridge as the model with the lowest test RMSE. The baseline Linear Regression model recorded RMSE 0.0679, MAE 0.0480, MAPE 8.51% and R^2 0.8212. Tuned Polynomial Ridge reduced RMSE to 0.0677, MAE to 0.0476 and MAPE to 8.44%, while increasing R^2 to 0.8224.

4.3.2 Percentage Improvement

Metric	Baseline	Improved	Change
RMSE	0.0679	0.0677	0.34% reduction
MAE	0.0480	0.0476	0.82% reduction
MAPE	8.51%	8.44%	0.89% reduction
R^2	0.8212	0.8224	0.15% increase

The improvement is deliberately reported as modest because the original Linear Regression baseline was already strong. The result demonstrates that polynomial expansion combined with Ridge regularization can capture a small amount of additional nonlinear structure without replacing the baseline with an unsupported claim of large improvement.

Chapter 5 – Standards Adopted

5.1 Design Standards

- Modular notebook structure with separate sections for loading, preprocessing, EDA, modelling and evaluation.
- Reproducible train-test splits using fixed random states.
- Pipeline-based preprocessing where appropriate to reduce leakage risk.
- Explicit separation of exploratory target-derived variables from predictive features.

5.2 Coding Standards

- Descriptive variable and model names.
- Comments and markdown explanations for major processing steps.
- Use of reusable evaluation functions and data transformations.
- Consistent numerical formatting for reported metrics.
- Warnings suppressed only where they do not affect the validity of the analysis.

5.3 Testing / Verification Standards

- Models are evaluated on held-out test data.
- Multiple metrics are reported rather than relying on a single score.
- Classification models are evaluated with confusion matrices and ROC-AUC.
- Project 3 compares all improvement methods against the same baseline and test set.
- Improvement percentages are calculated from observed test metrics only.

Chapter 6 – Conclusion and Future Scope

6.1 Conclusion

The Major Project successfully implemented three machine-learning studies covering regression, classification and accuracy improvement. Project 1 demonstrated an end-to-end housing-price prediction workflow and found Random Forest to have the lowest RMSE among the evaluated regressors. Project 2 demonstrated a loan-approval classification workflow and showed that Logistic Regression produced the highest accuracy, recall, F1 Score and ROC-AUC in the executed comparison, while Random Forest produced the highest precision. Project 3 established a strong Linear Regression baseline and achieved a small but measurable improvement using Tuned Polynomial Ridge.

The project also emphasized responsible evaluation. Target-derived price-per-square-foot information was excluded from the Project 1 predictive feature set, and artificial lag variables were not created for the cross-sectional Admission dataset. The results therefore provide a reproducible comparison based on the actual executed experiments.

6.2 Future Scope

- Evaluate the models on larger and more recent datasets.
- Use cross-validation and external validation datasets to test generalization more thoroughly.
- Perform systematic feature selection and explainability using methods such as permutation importance or SHAP.
- Investigate calibrated classification probabilities for loan-approval applications.
- Use genuine time-series Admission or institutional datasets if date-indexed observations become available, enabling valid lag, rolling-window and LSTM experiments.
- Deploy selected models through a lightweight API or dashboard for interactive prediction and monitoring.

References

- [1] Major Project – August 2026 Batch, Data Science & Business Analytics project guideline.
- [2] ProjectReport_Sample_, supplied project report sample.
- [3] Scikit-learn documentation for preprocessing, regression, classification, model selection and evaluation metrics.
- [4] Standard Bengaluru/Bangalore House Price dataset used for Project 1.
- [5] Standard Loan Prediction / Loan Eligibility dataset used for Project 2.
- [6] Standard Graduate Admissions / Admission_Predict.csv dataset used for Project 3.
- [7] C. M. Bishop, Pattern Recognition and Machine Learning, Springer.
- [8] I. Goodfellow, Y. Bengio and A. Courville, Deep Learning, MIT Press.

Individual Contribution

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The project work was carried out individually as part of the DSBA Major Project requirements. The contribution covered dataset preparation, exploratory data analysis, preprocessing, feature engineering, machine-learning model implementation, evaluation, comparison and documentation across all three project components.

For Project 1, the work included cleaning the housing dataset, parsing property-area values, creating engineered variables, handling outliers, training three regression algorithms and analysing feature importance. For Project 2, the work included categorical preprocessing, class-imbalance analysis, four classification algorithms, confusion-matrix/ROC evaluation and practical banking interpretation. For Project 3, the work included baseline establishment, feature engineering, advanced regression experiments, hyperparameter tuning and calculation of measurable improvement.

The implementation was executed in Python using Google Colab/Jupyter-compatible notebooks. The final results presented in this report correspond to the actual notebook executions completed during the project work.